```
--> //примеры работы с массивами

--> F=1:8
 F  =

   1.    2.    3.    4.    5.    6.    7.    8.

--> E=-5.2; L=7.2;

--> T=E:L
 T  =

         column 1 to 12

  -5.2  -4.2  -3.2  -2.2  -1.2  -0.2   0.8   1.8   2.8   3.8   4.8   5.8

         column 13

   6.8

--> 1:4
 ans  =

   1.    2.    3.    4.

--> //Определение вектора-строки

--> C=[12 6 7]
 C  =

   12.   6.   7.
```

```
--> M=[8 7 90 66]
 M  =

   8.    7.    90.    66.

--> L=[-6.2, 4.1, -8.3]
 L  =

  -6.2   4.1  -8.3

--> //Определение вектора-столбца

--> E=[5;2;6]
 E  =

   5.
   2.
   6.

--> O=[1;5;6;5;7]
 O  =

   1.
   5.
   6.
   5.
   7.

--> Q=[1]
 Q  =

   1.
```

```
--> //примеры обращения к элементу массива

--> X=[5,2];

--> X(2)+4*X(1)
 ans  =

   22.

--> Y=[3,9,8,6];

--> Y(2)+Y(4)/Y(1)
 ans  =

   11.

--> Z=[4;5;8;10;15;3];

--> Z(1)/Z(3)+2*Z(6)+Z(5)/Z(2)
 ans  =

   9.5

--> //примеры обращения к элементам матрицы

--> X=[5 2;3 9];

--> 3*X
 ans  =

   15.   6.
   9.    27.
```

```
--> Y=[8 6;7 4];

--> Y^2
 ans  =

   106.   72.
   84.    58.

--> Z=[40 12 6;21 36 1]
 Z  =

   40.   12.   6.
   21.   36.   1.

--> 2*Z
 ans  =

   80.   24.   12.
   42.   72.   2.

--> //пример конкатенации матриц

--> A1=[5 6 7];A2=[8 9 0];A3=[10 11 12];

--> A=[A1;A2;A3]
 A  =

   5.    6.    7.
   8.    9.    0.
   10.   11.   12.
```

```
--> B=[A A A]
 B  =

   5.    6.    7.    5.    6.    7.    5.    6.    7.
   8.    9.    0.    8.    9.    0.    8.    9.    0.
   10.   11.   12.   10.   11.   12.   10.   11.   12.

--> C=[A;A]
 C  =

   5.    6.    7.
   8.    9.    0.
   10.   11.   12.
   5.    6.    7.
   8.    9.    0.
   10.   11.   12.

--> //Примеры использования операции ":"

--> X=[1 2 3;4 5 6;7 8 9]
 X  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.

--> X(:,3)
 ans  =

   3.
   6.
   9.
```

```
--> Y=[5 9;8 6]
 Y  =

   5.   9.
   8.   6.

--> Y(2,:)=[]
 Y  =

   5.   9.

--> Y=M(:)
 Y  =

   8.
   7.
   90.
   66.

--> B=Y(1:2)
 B  =

   8.
   7.

--> //примеры матричных операций

--> A=[15 -7;1 20]
 A  =

   15.  -7.
   1.    20.
```

```
--> B=[1 5;20 0]
 B  =

   1.     5.
   20.    0.

--> A'+B
 ans  =

   16.    6.
   13.    20.

--> A=[-1 2;2 -3]
 A  =

  -1.    2.
   2.   -3.

--> B=[-2 3;1 -4]
 B  =

  -2.    3.
   1.   -4.

--> X=B/A
 X  =

   0.   -1.
  -5.   -2.
```

```
--> //пример применения функции к массиву

--> s=[1 0.5 sqrt(3)/2;-0.5 -1 1];

--> cos(s)
 ans  =

   0.5403023   0.8775826   0.6478593
   0.8775826   0.5403023   0.5403023

--> q=[%pi 1;-%pi/2 -1];

--> sin(q)
 ans  =

   1.225D-16   0.841471
  -1.         -0.841471

--> //использование функции matrix

--> A=[1 -1;-2 2];

--> matrix(A,2,2)
 ans  =

   1.  -1.
  -2.   2.
```

```
--> Q=[1 5 9;12 2 4];

--> matrix(Q,2,3)
 ans  =

   1.    5.   9.
   12.   2.   4.

--> O=[22 5 7 4 1;5 10 4 5 8;78 85 6 4 5];

--> matrix(O,5,3)
 ans  =

   22.   85.   5.
   5.    7.    4.
   78.   4.    1.
   5.    6.    8.
   10.   4.    5.

--> //использование функции ones

--> ones(5,5)
 ans  =

   1.   1.   1.   1.   1.
   1.   1.   1.   1.   1.
   1.   1.   1.   1.   1.
   1.   1.   1.   1.   1.
   1.   1.   1.   1.   1.
```

```
--> ones(3,6)
 ans  =

   1.   1.   1.   1.   1.   1.
   1.   1.   1.   1.   1.   1.
   1.   1.   1.   1.   1.   1.

--> ones(1,8)
 ans  =

   1.   1.   1.   1.   1.   1.   1.   1.

--> m=5; n=2;

--> D=ones(m,n)
 D  =

   1.   1.
   1.   1.
   1.   1.
   1.   1.
   1.   1.

--> //использование функции zeros

--> zeros(2,4)
 ans  =

   0.   0.   0.   0.
   0.   0.   0.   0.
```

```
--> K=[1 6 9 8];

--> L=zeros(K)
 L  =

   0.   0.   0.   0.

--> //примеры использования функции eye

--> eye(5,6)
 ans  =

   1.   0.   0.   0.   0.   0.
   0.   1.   0.   0.   0.   0.
   0.   0.   1.   0.   0.   0.
   0.   0.   0.   1.   0.   0.
   0.   0.   0.   0.   1.   0.

--> eye(4,4)
 ans  =

   1.   0.   0.   0.
   0.   1.   0.   0.
   0.   0.   1.   0.
   0.   0.   0.   1.

--> m=4;n=2;
```

```
--> E=eye(m,n)
 E  =

   1.   0.
   0.   1.
   0.   0.
   0.   0.

--> //примеры использования функции rand

--> A=[15 1;-7 20];

--> rand(A)
 ans  =

   0.2113249   0.0002211
   0.7560439   0.3303271

--> rand(5,5)
 ans  =

   0.6653811   0.068374    0.5442573   0.6525135   0.3616361
   0.6283918   0.5608486   0.2320748   0.3076091   0.2922267
   0.8497452   0.6623569   0.2312237   0.9329616   0.5664249
   0.685731    0.7263507   0.2164633   0.2146008   0.4826472
   0.8782165   0.1985144   0.8833888   0.312642    0.3321719
```

```
--> B=rand(3,3,3)
 B  =

(:,:,1)

   0.5935095   0.2693125   0.9184708
   0.5015342   0.6325745   0.0437334
   0.4368588   0.4051954   0.4818509
(:,:,2)

   0.2639556   0.1280058   0.1121355
   0.4148104   0.7783129   0.6856896
   0.2806498   0.211903    0.1531217
(:,:,3)

   0.6970851   0.4094825   0.1998338
   0.8415518   0.8784126   0.5618661
   0.4062025   0.113836    0.5896177

--> //использование функций sparse, full

--> A=sparse([1 2;3 4;5 6],[7 8 5])
 A  =

(  5,   6) sparse matrix

(  1,   2)         7.
(  3,   4)         8.
(  5,   6)         5.
```

```
--> B=sparse([9 8;5 6],[8 6])
 B  =

(  9,   8) sparse matrix

(  5,   6)          6.
(  9,   8)          8.

--> C=sparse([10 23;12 56;67 85],[52 41 63])
 C  =

(  67,   85) sparse matrix

(  10,   23)        52.
(  12,   56)        41.
(  67,   85)        63.

--> full(A)
 ans  =

   0.    7.    0.    0.    0.    0.
   0.    0.    0.    0.    0.    0.
   0.    0.    0.    8.    0.    0.
   0.    0.    0.    0.    0.    0.
   0.    0.    0.    0.    0.    5.
```

```
--> full(B)
 ans  =

   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   6.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   8.

--> //использование функции diag

--> A=[4 5 18 20];

--> diag(A)
 ans  =

   4.   0.   0.    0.
   0.   5.   0.    0.
   0.   0.   18.   0.
   0.   0.   0.    20.

--> diag(A,2)
 ans  =

   0.   0.   4.   0.   0.    0.
   0.   0.   0.   5.   0.    0.
   0.   0.   0.   0.   18.   0.
   0.   0.   0.   0.   0.    20.
   0.   0.   0.   0.   0.    0.
   0.   0.   0.   0.   0.    0.
```

```
--> diag(A,-2)
 ans  =

   0.    0.    0.     0.     0.    0.
   0.    0.    0.     0.     0.    0.
   4.    0.    0.     0.     0.    0.
   0.    5.    0.     0.     0.    0.
   0.    0.    18.    0.     0.    0.
   0.    0.    0.     20.    0.    0.

--> //использование функции cat

--> A=[1 2 6;12 0 5]; B=[52 4 1; 9 6 5];

--> cat(3,A,B)
 ans  =

(:,:,1)

   1.     2.    6.
   12.    0.    5.
(:,:,2)

   52.    4.    1.
   9.     6.    5.

--> cat(2,A,B)
 ans  =

   1.     2.    6.    52.    4.    1.
   12.    0.    5.    9.     6.    5.
```

```
--> //использование функции tril

--> A=[5 6 8 4;51 14 1 5;0 1 2 3;5 4 6 4]
 A  =

   5.    6.    8.    4.
   51.   14.   1.    5.
   0.    1.    2.    3.
   5.    4.    6.    4.

--> tril(A)
 ans  =

   5.    0.    0.    0.
   51.   14.   0.    0.
   0.    1.    2.    0.
   5.    4.    6.    4.

--> tril(A,2)
 ans  =

   5.    6.    8.    0.
   51.   14.   1.    5.
   0.    1.    2.    3.
   5.    4.    6.    4.

--> B=[1 4 5;1 5 6;1 2 4]
 B  =

   1.   4.   5.
   1.   5.   6.
   1.   2.   4.
```

```
--> tril(B)
 ans  =

   1.   0.   0.
   1.   5.   0.
   1.   2.   4.

--> tril(B,-2)
 ans  =

   0.   0.   0.
   0.   0.   0.
   1.   0.   0.

--> //использование функции triu

--> A=[1 2 3;4 5 6;7 8 9]
 A  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   9.

--> triu(A)
 ans  =

   1.   2.   3.
   0.   5.   6.
   0.   0.   9.
```

```
--> B=[0 9 8;7 6 5;4 3 2]
 B  =

   0.   9.   8.
   7.   6.   5.
   4.   3.   2.

--> triu(B,2)
 ans  =

   0.   0.   8.
   0.   0.   0.
   0.   0.   0.

--> //использование функции size

--> A=[1 5;23 5;4 5];

--> [n,m]=size(A)
 n  =

   3.
 m  =

   2.

--> //использование функции length

--> A=[1 5 23 6 -4];

--> length (A)
 ans  =

   5.
```

```
--> length (ans)
 ans  =

   1.

--> B=[1 4 52];

--> length (B)
 ans  =

   3.

--> //использование функции sum

--> A=[15 1 3 -6;-7 20 4 2];

--> X=sum(A)
 X  =

   32.

--> B=[9 5 7 6;-9 5 6 -20];

--> Y=sum(B,2)
 Y  =

   27.
  -18.

--> //использование функции prod
```

```
--> prod(A)
 ans  =

   302400.

--> p1=prod(B,2)
 p1  =

   1890.
   5400.

--> p2=prod(A,1)
 p2  =

  -105.   20.   12.  -12.

--> X=[1 5 9];

--> prod(X)
 ans  =

   45.

--> //использование функции max и min

--> A=[1 8 95 4;5 5 6 2;14 14 55 6;5 1 2 5];

--> max(A)
 ans  =

   95.
```

```
--> min(A)
 ans  =

   1.

--> B=[1 2 3;5 26 -5;-4 5 -6];

--> max(B)
 ans  =

   26.

--> min(B)
 ans  =

  -6.

--> C=[1 2;3 6];

--> max(C)
 ans  =

   6.

--> min(C)
 ans  =

   1.

--> //использование функции mean
```

```
--> mean(A)
 ans  =

   14.25

--> mean(B)
 ans  =

   3.

--> mean(C)
 ans  =

   3.

--> //использование функции median

--> A=[1 5 6 3;5 7 -2 -9;10 52 4 6;-5 -4 -6 -5];

--> median(A)
 ans  =

   3.5

--> B=[1 5;2 6];

--> median(B,1)
 ans  =

   1.5   5.5
```

```
--> //использование функций det, rank

--> X=[1 2 3;4 5 6;7 8 0];

--> det(X)
 ans  =

   27.000000

--> Y=[15 1;-7 20];

--> det(Y)
 ans  =

   307.

--> Z=[3 2 1;2 3 1;2 1 3];

--> det(Z)
 ans  =

   12.000000

--> rank(X)
 ans  =

   3.

--> rank(Y)
 ans  =

   2.
```

```
--> rank(Z)
 ans  =

   3.

--> //использование функции norm

--> X=[1 5 2;5 6 9;1 0 2];

--> norm(X,1)
 ans  =

   13.

--> Y=[4 5;2 3];

--> norm(Y,2)
 ans  =

   7.3434205

--> //остальные функции

--> X=[1 2 3;4 5 6;7 8 0]
 X  =

   1.   2.   3.
   4.   5.   6.
   7.   8.   0.
```

```
--> spec(X)
 ans  =

   12.122894 + 0.i
  -0.3883838 + 0.i
  -5.7345099 + 0.i

--> inv(X)
 ans  =

  -1.7777778   0.8888889  -0.1111111
   1.5555556  -0.7777778   0.2222222
  -0.1111111   0.2222222  -0.1111111

--> pinv(X)
 ans  =

  -1.7777778   0.8888889  -0.1111111
   1.5555556  -0.7777778   0.2222222
  -0.1111111   0.2222222  -0.1111111

--> Y=[1 2 3;3 2 1;2 3 1]
 Y  =

   1.   2.   3.
   3.   2.   1.
   2.   3.   1.

--> spec(Y)
 ans  =

   6. + 0.i
  -1. + i
  -1. - i
```

```
--> inv(Y)
 ans  =

  -0.0833333   0.5833333  -0.3333333
  -0.0833333  -0.4166667   0.6666667
   0.4166667   0.0833333  -0.3333333

--> pinv(Y)
 ans  =

  -0.0833333   0.5833333  -0.3333333
  -0.0833333  -0.4166667   0.6666667
   0.4166667   0.0833333  -0.3333333

--> rref(Y)
 ans  =

   1.   0.   0.
   0.   1.   0.
   0.   0.   1.

--> Z=[5 2 3;1 2 3;4 7 5]
 Z  =

   5.   2.   3.
   1.   2.   3.
   4.   7.   5.
```

M

```
--> rref(Z)
 ans  =

   1.   0.   0.
   0.   1.   0.
   0.   0.   1.

--> //формирование символьных матриц

--> X=['w' 'x' 'y' 'z']
 X  =

  "w"  "x"  "y"  "z"

--> Y=['4' '3' '2' '1']
 Y  =

  "4"  "3"  "2"  "1"

--> X+Y
 ans  =

  "w4"  "x3"  "y2"  "z1"

--> Y'
 ans  =

  "4"
  "3"
  "2"
  "1"
```